Rational Functions and Discontinuities

Concept

Rational graphs can have holes and asymptotes. Start with the original denominator.

Domain

Exclude every original denominator zero before cancelling factors.

Hole / vertical asymptote

A fully cancelled factor gives a hole; a remaining denominator zero gives an asymptote.

Horizontal asymptote

Lower numerator degree: $y=0$; equal degrees: leading-coefficient ratio.

Slant asymptote

If numerator degree is one larger, divide to find a linear quotient.

Example 1

Find the key features of $(x^2 - 1)/(x^2 - x - 2)$.

Solution

1. Factor and cancel $x+1$: $(x-1)/(x-2)$, with $x \neq -1,2$.
2. Hole at $(-1,2/3)$; vertical asymptote $x=2$; horizontal asymptote $y=1$.
3. Intercepts: $(1,0)$ and $(0,1/2)$.

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Example 2

Find the asymptotes of $y = (2x + 3)/(x - 1)$.

Solution

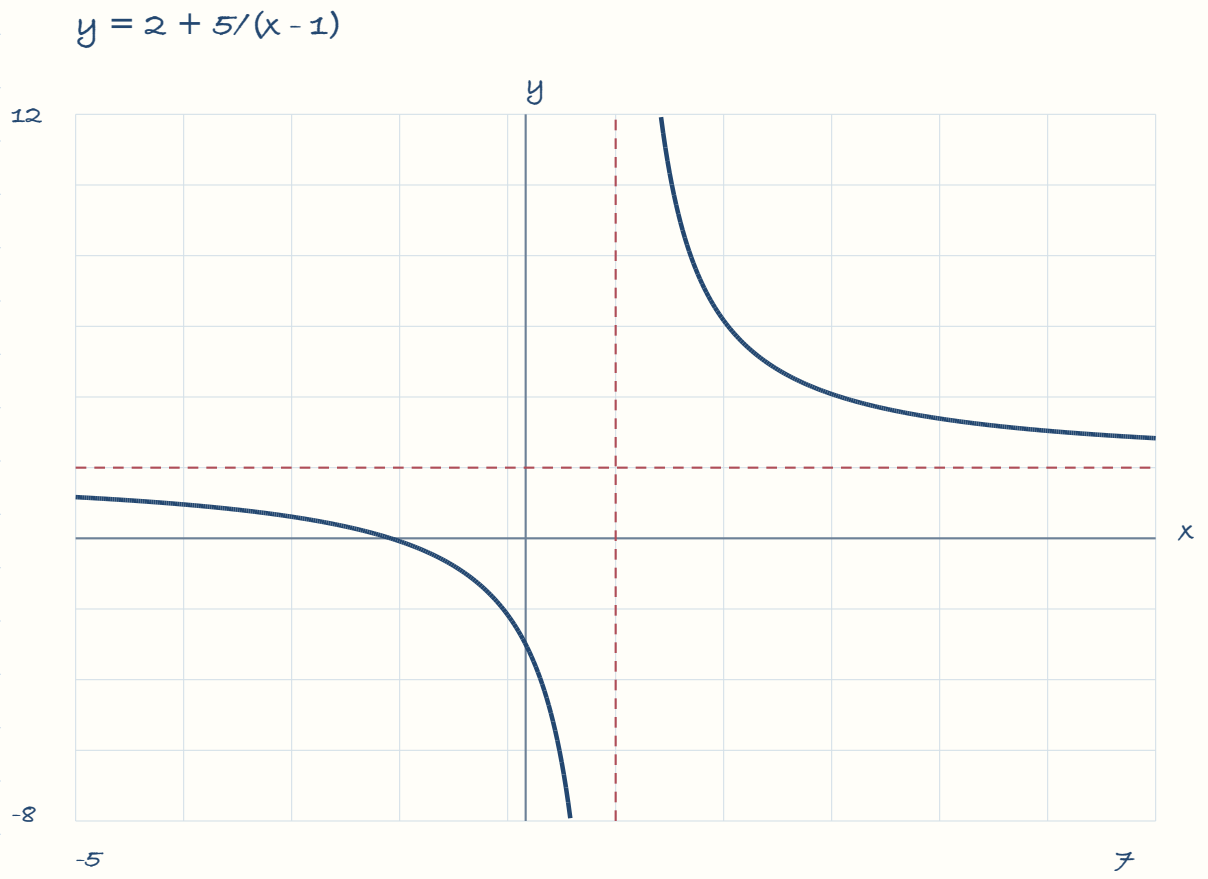
1. Rewrite: $y = 2 + 5/(x - 1)$.

2. Vertical asymptote $x = 1$; horizontal asymptote $y = 2$.

3. Domain $x \neq 1$; range $y \neq 2$.

Rational Functions and Discontinuities

Visual notes | read the graph



Dashed asymptotes: $x = 1$ and $y = 2$. Neither value is attained as the corresponding input or output.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Rational Functions and Discontinuities

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Describe the graph features of $(x^2 - 4)/(x - 2)$.

2. Find vertical and horizontal asymptotes of $(3x + 1)/(2x - 4)$.

3. Find intercepts of $(x - 3)/(x + 2)$.

4. Can $y = 1/(x^2 + 1)$ have a vertical asymptote over the reals?

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Practice answers | check after trying

1. Line $y = x + 2$ with hole $(2,4)$; domain $x \neq 2$. No vertical asymptote at 2 because the denominator factor cancels completely.

2. $x = 2$; $y = 3/2$.

3. $(3,0)$ and $(0,-3/2)$.

4. No. $x^2 + 1$ is positive for every real x, so the denominator never vanishes.